# Week 87: Voting and agendas

Adult guide built from the problem files · Unpiloted; cutting, token fit, resets and timing physically unrehearsed.

This view repeats, by problem, the answer keys of the print guide, week-87-facilitator.pdf, whose source is editable/guide/facilitator.tex; page references are to that guide. Change both together; check_guide_copy.py compares them.

## Mathematical overview

Three fixed sincere rankings $ABC$, $BCA$, $CAB$ give a majority cycle: $A$ beats $B$, $B$ beats $C$, and $C$ beats $A$, each 2–1. A vote chooses the higher of the two currently offered options on each card. Every individual ranking is consistent; the pairwise majority comparisons need not form a consistent ranking.

If two options vote first and their winner meets the waiting option, this same profile lets any option win by waiting. No card or ballot rule changes. Conversely, any option that beats both others wins every such three-option knockout. This finite example does not prove Arrow's theorem and makes no claim about fairness or real elections.

**Five fixed distance voters on one common line have a strict winner at the third favorite.** Order favorites $p_1 \leq \cdots \leq p_5$ with multiplicity and assume $p_3$ is available. Against any $q < p_3$, voters 3,4,5 strictly prefer $p_3$; against any $q > p_3$, voters 1,2,3 strictly prefer it. Equal-distance ties among the other voters cannot defeat these three votes. This proves a unique strict pairwise winner against every other available location. A favorite at zero is a voter. Repeated favorites are separate voters, not one position. The middle favorite need not be the mean or the midpoint of the printed line.

For four favorites 1,3,7,9 on the line, every point in $[3, 7]$ never loses a comparison, but no point beats every other. For the integer board these weak winners are 3,4,5,6,7. The even-voter result is different from saying every vote ties. With three distinct candidate positions on a line, the middle candidate cannot be any distance voter's strict worst choice. Since the cyclic profile puts each candidate last once, no ordering of candidate positions realizes all three cards by nearest-distance preferences.

## Problems and routes

| Problem | 2–5 | 3–5 | 4–5 | route |
|---|---|---|---|---|
| Is there a letter that beats both others? | P1 | | | first |
| Choose who waits | P2 | | | first |
| Design a fixed winner and a waiting-letter winner | P3 | | | first |
| A location that beats every other | | P1 | | first |
| Repeated and uneven favorites | | P2 | | first |
| Does the third favorite always win? | | P3 | | first |
| Four voters: never losing is not winning | | P4 | | return |
| Can the waiting letter change the winner? | | | P1 | return |
| Can one line produce the cycle? | | | P2 | return |

**First route.** Grades 2–5 P1–P3; Grades 3–5 P1–P3; Grades 4–5 none. **Return visits.** Grades 2–5 none; Grades 3–5 P4; Grades 4–5 P1–P2.

## Key, one entry per problem

### Is there a letter that beats both others?

Grades 2–5 P1 · first route

*Idea.* The cards ABC, BCA, CAB give a majority cycle: A beats B, B beats C and C beats A, each 2 to 1, though every card is a consistent ranking.

With cards $ABC$, $BCA$, $CAB$, V1 and V3 choose A over B; V1 and V2 choose B over C; V2 and V3 choose C over A. Every result is 2–1. Keep all three cards visible, including the unused option in each ranking. A voter's favorite being absent does not remove that voter: compare the two options that are actually offered. There is no overall top option in these pair results, even though each individual card has a strict first, second and third choice.

The non-task worked example uses $DFE$, $EFD$, $FED$. When D and E are offered, V1 chooses D and V2/V3 choose E, so E wins 2–1. The fixed cards remain unchanged in the next demonstration: E then meets waiting F; V1/V3 choose F and V2 chooses E, so F wins 2–1. A child who votes for an unavailable F in the first round needs the available pair made explicit, not a new preference card.

## Choose who waits

Grades 2–5 P2 · first route

*Idea.* On the cyclic cards the letter that waits always wins the knockout, so the agenda alone decides the winner.

Keep the Problem 1 cards. To make A win, first compare B with C; B wins, then A beats B. To make B win, first compare C with A; C wins, then B beats C. To make C win, first compare A with B; A wins, then C beats A. Thus the waiting option wins each time. These three choices exhaust all unordered first pairs. Reversing the left/right position of a pair changes no ballot under the rule.

Distinguish the first-round loser, first-round winner, waiting option and final winner. The knockout mat retains those states. Reset the movable options between attempts; do not reset or rewrite the preference cards. A child can record just the first pair and final winner. This is an agenda effect under sincere pairwise voting, not evidence that voters changed their minds or voted strategically.

## Design a fixed winner and a waiting-letter winner

Grades 2–5 P3 · first route

*Idea.* A letter that beats both others wins every knockout; with three different cards the waiting letter can matter only for ABC, BCA, CAB or its mirror ACB, CBA, BAC.

A suitable three-distinct-card profile with one winner for every agenda is $ABC$, $ACB$, $BAC$. A beats B 2–1 and C 3–0. If A waits, it beats the first winner; if A plays first, it wins both of its comparisons. The cards need not have identical first choices. The cyclic profile of Problem 1 is already a valid three-distinct-card agenda-dependent example; children may reproduce or relabel it, or invent another and test their claim.

Do not require every child to enumerate all possible profiles. Accept a construction with its checkable pair votes and agenda argument. For a held hint, ask what an option would need to do against each of the other two. Blank ranking cards and movable slips preserve each invented profile without repeatedly transcribing it. If two identical cards appear, remind the child of this task's distinct-card condition; a repeated ranking would still be a legitimate voter in the general model.

## A location that beats every other

Grades 3–5 P1 · first route

*Idea.* With five voters on a line, the third favorite beats every other location: three voters always defend it.

Location 4 beats every other integer location. Against 0, the voters at 4,7,10 choose 4; the voter at 0 chooses 0; the voter at 2 ties. The count is 3–1 with one Tie. Against 10 the voters at 0,2,4 choose 4; the voter at 10 chooses 10; the voter at 7 ties: again 3–1 with one Tie. Against each of 1,2,3,5,6,7,8,9, the count is 3–2. A tied individual ballot is not an overall tied result.

The middle favorite is the answer, but do not state that before children compare locations. If a group gets bogged down testing every pair, invite them to choose a promising proposal and let a partner try to defeat it. Save the side-coalition explanation for a discovered conjecture or Problem 3. Distinguish a successful challenge from a tied vote: a strict winner must win every comparison, so either result refutes that claim.

## Repeated and uneven favorites

Grades 3–5 P2 · first route

*Idea.* Repeated favorites are separate voters, and the winner is the middle favorite, not the midpoint or the mean.

For favorites 0,0,0,8,10, the winner is 0. Three different voters at zero each prefer zero to every distinct proposal. Their markers occupy different lanes and their votes remain separate. For 1,4,4,4,9, the winner is 4 for the same three-voter reason. For 0,1,6,9,10, the winner is 6. This last configuration distinguishes the middle favorite from the board midpoint 5 and arithmetic mean $26/5$.

Do not combine coincident favorites into one token or discard a voter at zero. These examples deliberately retain multiplicity. A held adult prompt is to point to each voter's identity and ask which one has not yet voted, without selecting their vote. A tally that contains fewer than five tokens is unfinished, even if the outcome is already forced.

## Does the third favorite always win?

Grades 3–5 P3 · first route

*Idea.* Median voter theorem on a line: sorted favorites p1 to p5, voters 3-5 defend p3 against anything to its left and voters 1-3 against anything to its right.

Sort the five favorites with repeats retained: $p_1 \le p_2 \le p_3 \le p_4 \le p_5$. The third, $p_3$, wins against every distinct proposal q. If q is left of $p_3$, each of voters 3,4,5 is at or right of $p_3$, so moving farther left to q strictly increases their distance. Those three vote for $p_3$. If q is right of $p_3$, voters 1,2,3 supply the analogous three votes. The remaining two votes, including any ties, cannot reverse the majority.

This proof works on a continuous line too, provided the median location is an available option. It does not prove a result for arbitrary rankings, a circle, two-dimensional distance, or a restricted menu excluding the median. Testing every five-favorite multiset on these eleven ticks checks 3,003 arrangements but is not the reason the general theorem holds. Children can demonstrate the two defending groups using favorite markers and a proposal on either side, without algebra.

## Four voters: never losing is not winning

Grades 3–5 P4 · return route

*Idea.* With four voters at 1, 3, 7, 9 every location in [3, 7] never loses, yet none beats all others.

Exactly the tick locations 3,4,5,6,7 never lose. No location strictly beats every other location. To see the first assertion, take x between 3 and 7. Against any proposal left of x, voters at 7 and 9 favor x, so x has at least two votes and cannot lose among four voters. Against any proposal right of x, voters at 1 and 3 defend x in the same way. Two distinct proposals both inside $[3, 7]$ split the four votes 2–2. Hence none of those locations strictly beats all others.

A tick below 3 loses to 3: the voters at 3,7,9 all prefer 3. A tick above 7 similarly loses to 7, supported by voters at 1,3,7. This excludes all outside ticks and proves completeness. The same argument gives the whole closed interval $[3, 7]$ if all real locations are available. There are still winning comparisons elsewhere; "no universal strict winner" does not mean every comparison ties.

Remove both the fifth favorite marker and its vote token. Changing only one creates a different, inconsistent procedure. Let children preserve several tied comparisons before asking for an explanation of all the weak winners.

## Can the waiting letter change the winner?

Grades 4–5 P1 · return route

*Idea.* With three voters and three options the pair results are either a ranking with a letter that beats both others, or a cycle in which the waiting letter wins.

The first profile is $ABC$, $ACB$, $BAC$. A beats B 2–1 and C 3–0, so A wins every agenda by the argument in shared Problem 3. The other pair has B beating C 2–1, but its result cannot remove A.

The second profile is $ACB$, $CBA$, $BAC$. B beats A, A beats C, and C beats B, each 2–1. Thus each waiting option can win: first A/C then waiting B; first B/C then waiting A; first A/B then waiting C. The task asks children to use pair votes to explain the possible agenda effect, not merely replay every knockout. In the three-option, odd-voter strict-ranking model, the pair comparisons form either a transitive order with a universal winner or a three-cycle. This observation should not be silently extended to larger menus, where more patterns occur.

## Can one line produce the cycle?

Grades 4–5 P2 · return route

*Idea.* On a line the middle candidate is never a voter's strict last choice, so the cyclic cards, which put each letter last once, cannot come from distances.

No. Put the three distinct candidate locations in order $\ell < m < r$. For any voter favorite p at or left of m, the rightmost candidate r is strictly farther away than m. For any p at or right of m, the leftmost candidate $\ell$ is strictly farther away than m. At least one extreme is therefore farther from p than the middle candidate; m cannot be that voter's strict worst choice. This holds for arbitrary real positions and favorites, including favorites outside the candidate interval.

The cards $ABC$, $BCA$, $CAB$ each put a different candidate last. Whichever candidate occupies the middle would have to be strictly last for one of the voters, contradicting the preceding fact. Coincident candidates or tied distances cannot rescue a strict ranking. The worked conversion on the student page shows how one fixed favorite produces a ranking; failed experimental placements alone would not prove impossibility. This gives a concrete limit of the nearest-location model, not a judgment about people's preferences.